

Effective Attacks and Provable Defenses for Website Fingerprinting

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Tao Wang, *University of Waterloo*; Xiang Cai, Rishab Nithyanand, and Rob Johnson, *Stony Brook University*; Ian Goldberg, *University of Waterloo*

Website fingerprinting attacks allow a local, passive eavesdropper to identify a user's web activity by leveraging packet sequence information. These attacks break the privacy expected by users of privacy technologies, including low-latency anonymity networks such as Tor. In this paper, we show a new attack that achieves significantly higher accuracy than previous attacks in the same field, further highlighting website fingerprinting as a genuine threat to web privacy. We test our attack under a large open-world experimental setting, where the client can visit pages that the attacker is not aware of. We found that our new attack is much more accurate than previous attempts, especially for an attacker monitoring a set of sites with low base incidence rate. We can correctly determine which of 100 monitored web pages a client is visiting (out of a significantly larger universe) at an 85% true positive rate with a false positive rate of 0.6%, compared to the best of 83% true positive rate with a false positive rate of 6% in previous work.

To defend against such attacks, we need provably effective defenses. We show how simulatable, deterministic defenses can be provably private, and we show that bandwidth overhead optimality can be achieved for these defenses by using a supersequence over anonymity sets of packet sequences. We design a new defense by approximating this optimal strategy and demonstrate that this new defense is able to defeat any attack at a lower cost on bandwidth than the previous best.

Tao Wang, University of Waterloo

Xiang Cai, Stony Brook University

Rishab Nithyanand, Stony Brook University

Rob Johnson, Stony Brook University

Ian Goldberg, University of Waterloo

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